

Mathematical Methods · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

Exam 1 ·

41 marks

reading 15 min · writing 60 min

Exam 2 ·

69 marks

reading 15 min · writing 120 min

State average on these questions

72 / 110 (65%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the easiest questions.

Sources

- [2023 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)

Exam 1 · technology free (41 marks, reading 15 min, writing 60 min)

Questions (technology free)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (4 marks)

2023 VCAA Exam 1 · Question 1 · [open in the official exam PDF \(p. 3\)](#) ↗

Quotient/product rule for $\frac{x^2-x}{e^x}$; product rule for $\sin(x)e^{2x}$ at $\frac{\pi}{4}$

a. Find and simplify $\frac{dy}{dx}$ for $y = \frac{x^2-x}{e^x}$ (2 marks)

b. $f'(\frac{\pi}{4})$ for $f(x) = \sin(x)e^{2x}$ (2 marks)

Question 2 (3 marks)

2023 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 4\)](#) ↗

Solve $e^{2x} - 12 = 4e^x$

Solve for x (3 marks)

Question 3 (4 marks)

2023 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

Sketch the hyperbola $f(x) = 2 - \frac{3}{x-1}$; solve $f(x) \leq 1$

a. Sketch with asymptotes and intercepts (3 marks)

b. Values of x with $f(x) \leq 1$ (1 mark)

Question 4 (4 marks)

2023 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 6\)](#) ↗

Evaluate $\int_0^{\pi/3} \sin x \, dx$; find every k in $(-3\pi, 2\pi)$ with $\int_k^{\pi/2} \cos x \, dx$ equal to it

a. $\int_0^{\pi/3} \sin(x) \, dx$ (1 mark)

b. All k with the two integrals equal (3 marks)

Question 5 (3 marks)

2024 VCAA Exam 1 · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Product rule for $e^x \cos(3x)$; derivative of $\log_e(x^3 - 3x + 2)$ at a point

a. Differentiate $y = e^x \cos(3x)$ (1 mark)

b. Find $f'(3)$ for $f(x) = \log_e(x^3 - 3x + 2)$ (2 marks)

Question 6 (3 marks)

2025 VCAA Exam 1 · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Product rule for $x^2 \cos x$; chain rule for a square-root function, evaluated at a point

a. Differentiate $y = x^2 \cos(x)$ (1 mark)

b. Gradient of the tangent to $f(x) = 6\sqrt{x+1} + 5$ at $x = 8$ (2 marks)

Question 7 (6 marks)

2025 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

$f(x) = 2 \cos(2x) + 1$ on $[0, 2\pi]$: range, solving $f(x) = 0$, sketching part of the graph

a. State the range (1 mark)

b. Solve $f(x) = 0$ (3 marks)

c. Sketch on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$, labelling endpoints (2 marks)

Question 8 (4 marks)

2025 VCAA Exam 1 · Question 4 · [open in the official exam PDF \(p. 5\)](#) ↗

Discrete distribution with an unknown k : showing $k = 10$ or 15 , then a probability and the mean

a. Show that $k = 10$ or $k = 15$ (2 marks)

b.i. $\Pr(X > 1)$ when $k = 15$ (1 mark)

b.ii. $E(X)$ when $k = 15$ (1 mark)

Question 9 (4 marks)

2025 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 6\)](#) ↗

Solve $e^{2x} - 8e^x + 7 = 0$; largest domain restriction giving an inverse

a. Solve $e^{2x} - 8e^x + 7 = 0$ (2 marks)

b. Largest a so that g restricted to $(-\infty, a]$ has an inverse (2 marks)

Question 10 (6 marks)

2025 VCAA Exam 1 · Question 7 · [open in the official exam PDF \(p. 8\)](#) ↗

Cubic $f(x) = x^3 - x^2 - 16x - 20$: verify a root, factorise, intercepts, then $f(x)(x + 2)$

a. Verify $x = 5$ is a solution of $f(x) = 0$ (1 mark)

b. Write $f(x) = (x + d)^2(x - 5)$ (2 marks)

c. Axial intercepts (1 mark)

d.i. Stationary point of inflection of $y = f(x)g(x)$, $g(x) = x + 2$ (1 mark)

d.ii. Values of x with $f(x)g(x) \geq 0$ (1 mark)

Exam 2 · CAS and a bound reference allowed (69 marks, reading 15 min, writing 120 min)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–E. One mark each.

Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Amplitude and period of $f(x) = -\frac{1}{2}\sin(3x + 2\pi)$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 3\)](#) ↗

Which power function has a horizontal tangent at the origin

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 4\)](#) ↗

Domain of the derivative of $\log_e \sqrt{1-x}$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 5\)](#) ↗

k with $\Pr(X < k) = 0.35$ for a piecewise-linear pdf

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 7\)](#) ↗

Matching two normal pdfs ($\mu = 100$, $\sigma = 20$ and $\mu = 80$, $\sigma = 10$) to a diagram

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 8\)](#) ↗

g given $f(x)g(x) = e^{(x-1)^2}$ with $f(x) = e^{x-1}$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 7 · 1 mark

2024 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Asymptote(s) of $y = \log_e(x+1) - 3$

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2024 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗

Find $g(2)$ given $g'(x) = x^3 - x$ and $g(0) = 5$

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark2024 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 2\)](#) ↗Conditional probability $\Pr(X < 4 \mid X > 1)$ from a table in terms of k Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2024 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗ $\int_b^c 2f(x) dx$ from two other integralsYour answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2024 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 3\)](#) ↗Minimum number of die rolls so that $\Pr(\text{at least one six}) > 0.95$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 4\)](#) ↗Where $h = f - g$ must have an x-intercept, from a table of valuesYour answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2024 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 4\)](#) ↗

Probability a student came by limousine given they had a photo taken

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2025 VCAA Exam 2 · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗Which circular function has range $[6, 12]$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2025 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 3\)](#) ↗Choosing the graph of $y = f(-x) + 2$ from a given graphYour answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2025 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 4\)](#) ↗

Which set of ordered pairs has an inverse function

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2025 VCAA Exam 2 · Section A · Question 7 · [open in the official exam PDF \(p. 5\)](#) ↗

Tracing a while-loop algorithm

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2025 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 5\)](#) ↗Sample size from an approximate 95% confidence interval $(0.248, 0.552)$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark2025 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 7\)](#) ↗

Greatest average rate of change of a share price over given intervals

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark2025 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 9\)](#) ↗Value of k for a piecewise sine/cosine pdfYour answer: ☐ A ☐ B ☐ C ☐ D

Section B – Extended-response questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (11 marks)2023 VCAA Exam 2 · Section B · Question 1 · [open in the official exam PDF \(p. 10\)](#) ↗Cubic $f(x) = x(x - 2)(x + 1)$: intercepts, stationary points, area with a line, matching $(x - a)(x - b)^2$ **a.** Coordinates of all axial intercepts (1 mark)**b.** Coordinates of the stationary points (exact) (2 marks)**c.i.** Solve $f(x) = x - 2$ (1 mark)**c.ii.** Integral expression for the total area between f and g (2 marks)**c.iii.** Total area (2 d.p.) (1 mark)

d. a and b with $(x - a)(x - b)^2 = f(x) + k$ (4 marks)

Question 2 (11 marks)

2024 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 14\)](#) ↗

Room-temperature models (hybrid linear and exponential) and a heater's power function: rates, equal temperatures, energy used

a. $f'(t)$ as a hybrid function (2 marks)

b. Average rate of change from $t = 0$ to $\frac{1}{2}$ (1 mark)

c.i. $g'(t)$ for $g(t) = 22 - 10e^{-6t}$ (1 mark)

c.ii. t with $g'(t) = 10$ (3 d.p.) (1 mark)

d. Time in $(0, 1)$ when f and g predict the same temperature (2 d.p.) (1 mark)

e. Time in $(0, 1)$ when the models differ most (2 d.p.) (1 mark)

f.i. Show $A = 1.2e^4$ using continuity at $t = 0.4$ (1 mark)

f.ii. Time until 0.5 kWh has been used (1 mark)

f.iii. Time until 1 kWh has been used (2 d.p.) (2 marks)

Question 3 (13 marks)

2025 VCAA Exam 2 · Section B · Question 1 · [open in the official exam PDF \(p. 14\)](#) ↗

Quartic $g(x) = 4x^3 - 3x^4$: stationary points, sketch, gradient table, average value, transformations, link to a binomial

a. Coordinates of both stationary points (2 marks)

b. Sketch with stationary points and intercepts (2 marks)

c. Gradient table showing a stationary point of inflection (2 marks)

d. Average value of g on $[0, 2]$ (2 marks)

e. Three transformations taking g to h with a stationary inflection at $(1, 0)$ and a maximum at $(-1, 1)$ (3 marks)

f. Show $\Pr(X \geq 3) = g(p)$ for $X \sim \text{Bi}(4, p)$ (2 marks)

Question 4 (14 marks)

2025 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 20\)](#) ↗

Commute times: a polynomial pdf (mean, sd, lateness), binomial weeks, a normal wait time, three traffic lights

a.i. Mean commute time (1 mark)

a.ii. Standard deviation (2 marks)

b.i. Integral showing $\Pr(\text{late}) = 0.08704$ when $k = 47$ (1 mark)

b.ii. $\Pr(\text{late at least once in 5 days})$ (4 d.p.) (2 marks)

b.iii. $\Pr(0.4 \leq \hat{P} \leq 0.6)$ (4 d.p.) (2 marks)

b.iv. Integer k making $\Pr(\text{late at least once in a week}) = 0.2$ (1 d.p.) (2 marks)

c.i. $\Pr(\text{wait} < 3.5)$ with $\mu = 2.5$, $\sigma = 0.6$ (2 d.p.) (1 mark)

c.ii. σ giving a 2% chance of waiting longer than 3.5 min (2 d.p.) (1 mark)

d. Probability distribution of Y , the number of red lights (2 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide · Exam 1

Questions (technology free)

Question 1 (4 marks) · state average 2.9 / 4 (73%)

2023 VCAA Exam 1 · Question 1

a. Find and simplify $\frac{dy}{dx}$ for $y = \frac{x^2 - x}{e^x}$

☐ Correct use of the product or quotient rule

☐ $\frac{dy}{dx} = \frac{-x^2 + 3x - 1}{e^x}$

Answer / notes: $(2x - 1)e^{-x} - (x^2 - x)e^{-x} = -(x^2 - 3x + 1)e^{-x}$. Simplify fully.

State average 1.3 / 2 (65%); 42% scored full marks. Mark distribution: 0: 8% · 1: 50% · 2: 42%

Your mark: _____ / 2

b. $f'(\frac{\pi}{4})$ for $f(x) = \sin(x)e^{2x}$

☐ $f'(x) = e^{2x}(\cos x + 2 \sin x)$

☐ $f'(\frac{\pi}{4}) = \frac{3\sqrt{2}}{2}e^{\pi/2}$

Answer / notes: $e^{\pi/2}(\frac{\sqrt{2}}{2} + \sqrt{2}) = \frac{3\sqrt{2}}{2}e^{\pi/2}$

State average 1.6 / 2 (80%); 65% scored full marks. Mark distribution: 0: 9% · 1: 26% · 2: 65%

Your mark: _____ / 2

Question 2 (3 marks) · state average 2.0 / 3 (67%)

2023 VCAA Exam 1 · Question 2

Solve for x

☐ Rearranges: $e^{2x} - 4e^x - 12 = 0$

☐ $(e^x - 6)(e^x + 2) = 0$

☐ $x = \log_e 6$ (since $e^x > 0$)

Answer / notes: $x = \log_e 6$. Reject $e^x = -2$.

State average 2.0 / 3 (67%); 56% scored full marks. Mark distribution: 0: 29% · 1: 2% · 2: 13% · 3: 56%

Your mark: _____ / 3

Question 3 (4 marks) · state average 2.8 / 4 (70%)

2023 VCAA Exam 1 · Question 3

a. Sketch with asymptotes and intercepts

☐ Asymptotes $x = 1$ and $y = 2$

☐ Intercepts $(0, 5)$ and $(\frac{5}{2}, 0)$

☐ Correct shape (branches upper-left and lower-right of the asymptotes)

Answer / notes: Asymptotes $x = 1$, $y = 2$. It passes through $(0, 5)$ and $(\frac{5}{2}, 0)$.

State average 2.4 / 3 (80%); 66% scored full marks. Mark distribution: 0: 7% · 1: 7% · 2: 19% · 3: 66%

Your mark: _____ / 3

b. Values of x with $f(x) \leq 1$

☐ $1 < x \leq 4$

Answer / notes: Use the graph: $f(4) = 1$, and the graph is below 1 on $(1, 4]$.

State average 0.4 / 1 (40%); 38% scored full marks. Mark distribution: 0: 62% · 1: 38%

Your mark: _____ / 1

Question 4 (4 marks) · state average 2.5 / 4 (62%)

2023 VCAA Exam 1 · Question 5

a. $\int_0^{\pi/3} \sin(x) dx$

☐ $\frac{1}{2}$

Answer / notes: $[-\cos x]_0^{\pi/3} = \frac{1}{2}$

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: _____ / 1

b. All k with the two integrals equal

☐ $1 - \sin k = \frac{1}{2}$, so $\sin k = \frac{1}{2}$

☐ Finds the general/multiple solutions

☐ $k = -\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

Answer / notes: Four values in $(-3\pi, 2\pi)$: $-\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$.

State average 1.8 / 3 (60%); 35% scored full marks. Mark distribution: 0: 22% · 1: 17% · 2: 27% · 3: 35%

Your mark: _____ / 3

Question 5 (3 marks) · state average 2.0 / 3 (67%)

2024 VCAA Exam 1 · Question 1

a. Differentiate $y = e^x \cos(3x)$

☐ $e^x \cos(3x) - 3e^x \sin(3x)$

Answer / notes: $e^x (\cos(3x) - 3 \sin(3x))$

State average 0.8 / 1 (80%); 82% scored full marks. Mark distribution: 0: 18% · 1: 82%

Your mark: _____ / 1

b. Find $f'(3)$ for $f(x) = \log_e(x^3 - 3x + 2)$

☐ $f'(x) = \frac{3x^2 - 3}{x^3 - 3x + 2}$

☐ $f'(3) = \frac{24}{20} = \frac{6}{5}$

Answer / notes: $\frac{6}{5}$

State average 1.2 / 2 (60%); 54% scored full marks. Mark distribution: 0: 31% · 1: 15% · 2: 54%

Your mark: _____ / 2

Question 6 (3 marks) · state average 2.4 / 3 (80%)

2025 VCAA Exam 1 · Question 1

a. Differentiate $y = x^2 \cos(x)$

☐ $\frac{dy}{dx} = 2x \cos(x) - x^2 \sin(x)$

Answer / notes: $2x \cos(x) - x^2 \sin(x)$. Use brackets and tidy the signs.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: _____ / 1

b. Gradient of the tangent to $f(x) = 6\sqrt{x+1} + 5$ at $x = 8$

☐ $f'(x) = \frac{3}{\sqrt{x+1}}$

☐ $f'(8) = 1$

Answer / notes: $f'(8) = \frac{3}{\sqrt{9}} = 1$.

State average 1.5 / 2 (75%); 61% scored full marks. Mark distribution: 0: 18% · 1: 21% · 2: 61%

Your mark: _____ / 2

Question 7 (6 marks) · state average 3.8 / 6 (63%)

2025 VCAA Exam 1 · Question 3

a. State the range

☐ $[-1, 3]$

Answer / notes: $[-1, 3]$

State average 0.8 / 1 (80%); 81% scored full marks. Mark distribution: 0: 19% · 1: 81%

Your mark: _____ / 1

b. Solve $f(x) = 0$

☐ $\cos(2x) = -\frac{1}{2}$

☐ Considers $2x \in [0, 4\pi]$ (four solutions)

☐ $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

Answer / notes: $2x = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$, so $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

State average 2.0 / 3 (67%); 51% scored full marks. Mark distribution: 0: 18% · 1: 13% · 2: 18% · 3: 51%

Your mark: _____ / 3

c. Sketch on $[\frac{\pi}{2}, \frac{3\pi}{2}]$, labelling endpoints

☐ Correct shape: one full cycle rising to a maximum of 3 at $x = \pi$

☐ Endpoints $(\frac{\pi}{2}, -1)$ and $(\frac{3\pi}{2}, -1)$ labelled

Answer / notes: Starts at $(\frac{\pi}{2}, -1)$, peaks at $(\pi, 3)$, ends at $(\frac{3\pi}{2}, -1)$.

State average 1.0 / 2 (50%); 33% scored full marks. Mark distribution: 0: 34% · 1: 32% · 2: 33%

Your mark: _____ / 2

Question 8 (4 marks) · state average 2.8 / 4 (70%)

2025 VCAA Exam 1 · Question 4

a. Show that $k = 10$ or $k = 15$

☐ Sum of probabilities = 1: $\frac{6}{k} + \frac{3k}{75} = 1$

☐ $k^2 - 25k + 150 = 0$, so $(k - 10)(k - 15) = 0$

Answer / notes: $\frac{6}{k} + \frac{k}{25} = 1 \Rightarrow k^2 - 25k + 150 = 0$. Show every line in a “show that”.

State average 1.3 / 2 (65%); 56% scored full marks. Mark distribution: 0: 26% · 1: 18% · 2: 56%

Your mark: _____ / 2

b.i. $\Pr(X > 1)$ when $k = 15$

☐ $\frac{1}{3}$

Answer / notes: $\frac{15}{75} + \frac{2}{15} = \frac{1}{3}$

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: _____ / 1

b.ii. $E(X)$ when $k = 15$

☐ $\frac{6}{5}$

Answer / notes: $0 + \frac{30}{75} + 2 \cdot \frac{15}{75} + 3 \cdot \frac{2}{15} = \frac{6}{5}$

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 29% · 1: 71%

Your mark: _____ / 1

Question 9 (4 marks) · state average 2.5 / 4 (62%)

2025 VCAA Exam 1 · Question 5

a. Solve $e^{2x} - 8e^x + 7 = 0$

☐ Factorises: $(e^x - 7)(e^x - 1) = 0$

☐ $x = 0$ or $x = \log_e 7$

Answer / notes: Treat as a quadratic in e^x : $x = 0, \log_e 7$.

State average 1.5 / 2 (75%); 61% scored full marks. Mark distribution: 0: 14% · 1: 25% · 2: 61%

Your mark: _____ / 2

b. Largest a so that g restricted to $(-\infty, a]$ has an inverse

☐ $g'(x) = 2e^{2x} - 8e^x = 2e^x(e^x - 4)$

☐ $a = \log_e 4$

Answer / notes: The stationary point is at $x = \log_e 4$, and g is one-to-one to the left of it, so $a = \log_e 4$.

State average 1.0 / 2 (50%); 43% scored full marks. Mark distribution: 0: 42% · 1: 16% · 2: 43%

Your mark: _____ / 2

Question 10 (6 marks) · state average 4.1 / 6 (68%)

a. Verify $x = 5$ is a solution of $f(x) = 0$

☐ $f(5) = 125 - 25 - 80 - 20 = 0$

Answer / notes: $f(5) = 0$.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: _____ / 1

b. Write $f(x) = (x + d)^2(x - 5)$

☐ Divides by $(x - 5)$ or compares coefficients: $x^2 + 4x + 4$

☐ $d = 2$

Answer / notes: $f(x) = (x + 2)^2(x - 5)$

State average 1.6 / 2 (80%); 76% scored full marks. Mark distribution: 0: 17% · 1: 8% · 2: 76%

Your mark: _____ / 2

c. Axial intercepts

☐ $(-2, 0), (5, 0), (0, -20)$

Answer / notes: $(-2, 0), (5, 0), (0, -20)$

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

Your mark: _____ / 1

d.i. Stationary point of inflection of $y = f(x)g(x)$, $g(x) = x + 2$

☐ $(-2, 0)$

Answer / notes: $f(x)g(x) = (x + 2)^3(x - 5)$, so the triple root gives a stationary inflection at $(-2, 0)$.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: _____ / 1

d.ii. Values of x with $f(x)g(x) \geq 0$

☐ $x \leq -2$ or $x \geq 5$

Answer / notes: $x \in (-\infty, -2] \cup [5, \infty)$

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: _____ / 1

Marking guide · Exam 2

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	E	Amplitude is always positive: $\frac{1}{2}$. Period = $\frac{2\pi}{3}$. <i>Most popular wrong answer: B (16%).</i>	77% correct A 2% · B 16% · C 2% · D 3% · E 77%
☐	2	D	$y = x^{4/3}$ has $y' = \frac{4}{3}x^{1/3} = 0$ at $x = 0$. Powers below 1 give vertical tangents or cusps there. <i>Most popular wrong answer: B (16%).</i>	62% correct A 9% · B 16% · C 8% · D 62% · E 4%
☐	3	C	The composite exists for $x < 1$, and its derivative $-\frac{1}{2(1-x)}$ exists on the same set: $(-\infty, 1)$. <i>Most popular wrong answer: E (17%).</i>	60% correct A 3% · B 12% · C 60% · D 7% · E 17%
☐	4	B	$\frac{(k-1)^2}{40} = 0.35 \Rightarrow k = 1 + \sqrt{14} \approx 4.7$, which lies in the first piece.	60% correct A 8% · B 60% · C 14% · D 13% · E 5%
☐	5	A	The smaller σ gives a taller, narrower curve, and the smaller mean sits further left. <i>Most popular wrong answer: C (17%).</i>	65% correct A 65% · B 6% · C 17% · D 4% · E 6%
☐	6	B	$g = e^{(x-1)^2 - (x-1)} = e^{(x-1)(x-2)}$.	69% correct A 14% · B 69% · C 8% · D 5% · E 4%
☐	7	A	A log graph has only a vertical asymptote: $x = -1$. The -3 shifts the graph but adds no horizontal asymptote. <i>Most popular wrong answer: D (20%).</i>	74% correct A 74% · B 2% · C 3% · D 20%
☐	8	D	$g(x) = \frac{x^4}{4} - \frac{x^2}{2} + 5$, so $g(2) = 4 - 2 + 5 = 7$.	81% correct A 10% · B 4% · C 5% · D 81%
☐	9	C	$13k = 1$, then $\frac{\Pr(X=2,3)}{\Pr(X=2,3,4)} = \frac{8k}{10k} = \frac{4}{5}$.	67% correct A 7% · B 11% · C 67% · D 14%
☐	10	B	$\int_b^c f = \int_a^c f - \int_a^b f = 3 - (-5) = 8$, and doubling gives 16.	70% correct A 7% · B 70% · C 12% · D 9%
☐	11	C	$1 - \left(\frac{5}{6}\right)^n > 0.95 \Rightarrow n > 16.4$, so $n = 17$. <i>Most popular wrong answer: B (15%).</i>	68% correct A 7% · B 15% · C 68% · D 9%
☐	12	A	$h(1) = -3$ and $h(2) = 0$, so $(2, 0)$ is an intercept. Look for where $f = g$ in the table. <i>Most popular wrong answer: C (16%).</i>	65% correct A 65% · B 15% · C 16% · D 3%
☐	13	C	$\frac{0.45 \times 0.3}{0.45 \times 0.3 + 0.55 \times 0.6} = \frac{0.135}{0.465} = \frac{9}{31}$ <i>Most popular wrong answer: B (18%).</i>	66% correct A 4% · B 18% · C 66% · D 11%
☐	14	C	Range of $c \pm a \cos(\dots)$ is $[c - a , c + a]$. You need centre 9 and amplitude 3: $9 - 3 \cos(6x)$.	91% correct A 2% · B 5% · C 91% · D 2%
☐	15	B	Reflect in the y-axis, then shift up 2. Check a known point, e.g. the y-intercept moves from 1 to 3.	71% correct A 4% · B 71% · C 12% · D 13%

✓	Q	Ans	Why	How Victoria went
☐	16	B	It needs to be one-to-one: no repeated x-values (to be a function) and no repeated y-values. Only one set qualifies.	76% correct A 6% · B 76% · C 13% · D 5%
☐	17	C	$n = 17 \rightarrow 12 \rightarrow 7 \rightarrow 2$. Each is printed, and the loop stops once $n \leq 5$. <i>Most popular wrong answer: B (15%).</i>	72% correct A 9% · B 15% · C 72% · D 3%
☐	18	C	$\hat{p} = 0.4$, margin $0.152 = 1.96\sqrt{\frac{0.24}{n}}$, so $n \approx 40$. <i>Most popular wrong answer: B (18%).</i>	66% correct A 7% · B 18% · C 66% · D 8%
☐	19	D	Average rate = change in price $\div$ days. Compute the gradient of the chord for each interval, not just the biggest rise. <i>Most popular wrong answer: C (16%).</i>	75% correct A 4% · B 4% · C 16% · D 75%
☐	20	B	$k(1 - \frac{\sqrt{2}}{2}) + k(1 - \frac{\sqrt{2}}{2}) = 1 \Rightarrow k = \frac{1}{2-\sqrt{2}}$. <i>Most popular wrong answer: C (17%).</i>	66% correct A 12% · B 66% · C 17% · D 5%

Multiple-choice score: _____ / 20

Section B – Extended-response questions

Question 1 (11 marks) · state average 6.4 / 11 (58%)

2023 VCAA Exam 2 · Section B · Question 1

a. Coordinates of all axial intercepts

☐ $(-1, 0), (0, 0), (2, 0)$

Answer / notes: $(-1, 0), (0, 0), (2, 0)$. The origin is also the y-intercept.

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

Your mark: _____ / 1

b. Coordinates of the stationary points (exact)

☐ $f'(x) = 3x^2 - 2x - 2 = 0 \Rightarrow x = \frac{1 \pm \sqrt{7}}{3}$

☐ $\left(\frac{1-\sqrt{7}}{3}, \frac{14\sqrt{7}-20}{27}\right)$ and $\left(\frac{1+\sqrt{7}}{3}, \frac{-14\sqrt{7}-20}{27}\right)$

Answer / notes: Give exact y-values as well as x-values.

State average 1.6 / 2 (80%); 68% scored full marks. Mark distribution: 0: 6% · 1: 25% · 2: 68%

Your mark: _____ / 2

c.i. Solve $f(x) = x - 2$

☐ $x = 2, \frac{-1 \pm \sqrt{5}}{2}$

Answer / notes: $(x - 2)(x^2 + x - 1) = 0$

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: _____ / 1

c.ii. Integral expression for the total area between f and g

☐ $\int_{\frac{-1-\sqrt{5}}{2}}^{\frac{-1+\sqrt{5}}{2}} (f - g) dx$

☐ $+ \int_{\frac{-1+\sqrt{5}}{2}}^2 (g - f) dx$

Answer / notes: Split at the middle intersection, with the upper function first in each.

State average 1.4 / 2 (70%); 61% scored full marks. Mark distribution: 0: 25% · 1: 15% · 2: 61%

Your mark: _____ / 2

c.iii. Total area (2 d.p.)

☐ ≈ 5.95

Answer / notes: ≈ 5.95

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: _____ / 1

d. a and b with $(x - a)(x - b)^2 = f(x) + k$

☐ Expands and equates coefficients: $a + 2b = 1$, $2ab + b^2 = -2$

☐ $3b^2 - 2b - 2 = 0$, so $b = \frac{1 \pm \sqrt{7}}{3}$

☐ Matching $a = 1 - 2b$

☐ $(a, b) = \left(\frac{1-2\sqrt{7}}{3}, \frac{1+\sqrt{7}}{3}\right)$ or $\left(\frac{1+2\sqrt{7}}{3}, \frac{1-\sqrt{7}}{3}\right)$

Answer / notes: The repeated root b must sit at a turning point of f, so b is a stationary x-value. Equating coefficients is cleanest. The state averaged 1/4.

State average 1.0 / 4 (25%); 13% scored full marks. Mark distribution: 0: 61% · 1: 11% · 2: 9% · 3: 6% · 4: 13%

Your mark: _____ / 4

Question 2 (11 marks) · state average 6.7 / 11 (61%)

2024 VCAA Exam 2 · Section B · Question 2

a. $f'(t)$ as a hybrid function

☐ $f'(t) = 30$ for $0 < t < \frac{1}{3}$

☐ $f'(t) = 0$ for $t > \frac{1}{3}$

Answer / notes: 30 on $(0, \frac{1}{3})$ and 0 on $(\frac{1}{3}, \infty)$. f isn't differentiable at the join, so use open intervals.

State average 1.4 / 2 (70%); 48% scored full marks. Mark distribution: 0: 11% · 1: 41% · 2: 48%

Your mark: _____ / 2

b. Average rate of change from $t = 0$ to $\frac{1}{2}$

☐ $20^\circ\text{C per hour}$

Answer / notes: $\frac{22-12}{0.5} = 20^\circ\text{C/h}$

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: _____ / 1

c.i. $g'(t)$ for $g(t) = 22 - 10e^{-6t}$

☐ $60e^{-6t}$

Answer / notes: $g'(t) = 60e^{-6t}$

State average 1.0 / 1 (100%); 95% scored full marks. Mark distribution: 0: 5% · 1: 95%

Your mark: _____ / 1

c.ii. t with $g'(t) = 10$ (3 d.p.)

☐ $t = \frac{\log_e 6}{6} \approx 0.299$

Answer / notes: ≈ 0.299 h

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: _____ / 1

d. Time in (0, 1) when f and g predict the same temperature (2 d.p.)

☐ $t \approx 0.27$

Answer / notes: Solve $12 + 30t = 22 - 10e^{-6t}$: $t \approx 0.27$.

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

e. Time in (0, 1) when the models differ most (2 d.p.)

☐ $t \approx 0.12$

Answer / notes: Maximise $g - f$ on the first piece: $60e^{-6t} = 30 \Rightarrow t = \frac{\log_e 2}{6} \approx 0.12$. This beats the gap at $t = \frac{1}{3}$.

State average 0.3 / 1 (30%); 29% scored full marks. Mark distribution: 0: 71% · 1: 29%

Your mark: _____ / 1

f.i. Show $A = 1.2e^4$ using continuity at $t = 0.4$

☐ $0.3 + Ae^{-4} = 1.5 \Rightarrow A = 1.2e^4$

Answer / notes: Set the two pieces equal at $t = 0.4$.

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

f.ii. Time until 0.5 kWh has been used

☐ $\frac{1}{3}$ hour

Answer / notes: $1.5t = 0.5 \Rightarrow t = \frac{1}{3} \leq 0.4$

State average 0.4 / 1 (40%); 37% scored full marks. Mark distribution: 0: 63% · 1: 37%

Your mark: _____ / 1

f.iii. Time until 1 kWh has been used (2 d.p.)

☐ $0.6 + \int_{0.4}^T (0.3 + 1.2e^4 e^{-10t}) dt = 1$

☐ $T \approx 1.33$ hours

Answer / notes: The first 0.4 h uses 0.6 kWh. Solve for the rest: $T \approx 1.33$ h.

State average 0.7 / 2 (35%); 27% scored full marks. Mark distribution: 0: 61% · 1: 11% · 2: 27%

Your mark: _____ / 2

Question 3 (13 marks) · state average 8.5 / 13 (65%)

2025 VCAA Exam 2 · Section B · Question 1

a. Coordinates of both stationary points

☐ $g'(x) = 12x^2(1 - x) = 0$

☐ $(0, 0)$ and $(1, 1)$

Answer / notes: $(0, 0)$ and $(1, 1)$

State average 1.9 / 2 (95%); 91% scored full marks. Mark distribution: 0: 2% · 1: 7% · 2: 91%

Your mark: _____ / 2

b. Sketch with stationary points and intercepts

☐ Correct shape: stationary inflection at O, maximum at $(1, 1)$

☐ x-intercepts $(0, 0)$ and $(\frac{4}{3}, 0)$ labelled

Answer / notes: Flat inflection at the origin, a peak at $(1, 1)$, then down through $(\frac{4}{3}, 0)$.

State average 1.6 / 2 (80%); 62% scored full marks. Mark distribution: 0: 7% · 1: 31% · 2: 62%

Your mark: _____ / 2

c. Gradient table showing a stationary point of inflection

☐ x-values either side of 0 (e.g. $-1, 0, \frac{1}{2}$)

☐ g' : positive, 0, positive

Answer / notes: $g'(-1) = 24 > 0$, $g'(0) = 0$, $g'(\frac{1}{2}) = 1.5 > 0$. There is no sign change.

State average 1.3 / 2 (65%); 55% scored full marks. Mark distribution: 0: 26% · 1: 19% · 2: 55%

Your mark: _____ / 2

d. Average value of g on $[0, 2]$

☐ $\frac{1}{2} \int_0^2 (4x^3 - 3x^4) dx$

☐ $= -\frac{8}{5}$

Answer / notes: $\frac{1}{2} (16 - \frac{96}{5}) = -\frac{8}{5}$

State average 1.5 / 2 (75%); 72% scored full marks. Mark distribution: 0: 25% · 1: 4% · 2: 72%

Your mark: _____ / 2

e. Three transformations taking g to h with a stationary inflection at (1, 0) and a maximum at (-1, 1)

- ☐ Reflection in the y-axis
- ☐ Dilation by a factor of 2 from the y-axis
- ☐ Translation of 1 unit in the positive x-direction

Answer / notes: Reflect: max at (-1, 1). Dilate $\times 2$ horizontally: (-2, 1). Shift right 1: max (-1, 1), inflection (1, 0). Other correct orders also work.

State average 1.2 / 3 (40%); 20% scored full marks. Mark distribution: 0: 30% · 1: 41% · 2: 9% · 3: 20%

Your mark: _____ / 3

f. Show $\Pr(X \geq 3) = g(p)$ for $X \sim \text{Bi}(4, p)$

- ☐ $\binom{4}{3}p^3(1-p) + p^4$
- ☐ $= 4p^3 - 3p^4 = g(p)$

Answer / notes: $4p^3 - 4p^4 + p^4 = 4p^3 - 3p^4$

State average 1.0 / 2 (50%); 45% scored full marks. Mark distribution: 0: 44% · 1: 11% · 2: 45%

Your mark: _____ / 2

Question 4 (14 marks) · state average 8.3 / 14 (59%)

2025 VCAA Exam 2 · Section B · Question 3

a.i. Mean commute time

- ☐ $E(T) = 39$ minutes

Answer / notes: 39 minutes.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

Your mark: _____ / 1

a.ii. Standard deviation

- ☐ Uses $\text{var}(T) = E(T^2) - [E(T)]^2$
- ☐ $\text{sd} \approx 5.35$ min ($\sqrt{200/7}$)

Answer / notes: ≈ 5.35 minutes.

State average 1.5 / 2 (75%); 65% scored full marks. Mark distribution: 0: 17% · 1: 18% · 2: 65%

Your mark: _____ / 2

b.i. Integral showing $\Pr(\text{late}) = 0.08704$ when $k = 47$

- ☐ $\int_{47}^{59} f(t) dt = 0.08704$

Answer / notes: $\Pr(T > 47) = \int_{47}^{59} f(t) dt$

State average 0.8 / 1 (80%); 81% scored full marks. Mark distribution: 0: 19% · 1: 81%

Your mark: _____ / 1

b.ii. $\Pr(\text{late at least once in 5 days})$ (4 d.p.)

- ☐ $1 - (1 - 0.08704)^5$
- ☐ ≈ 0.3658

Answer / notes: ≈ 0.3658

State average 1.3 / 2 (65%); 62% scored full marks. Mark distribution: 0: 28% · 1: 10% · 2: 62%

Your mark: _____ / 2

b.iii. $\Pr(0.4 \leq \hat{P} \leq 0.6)$ (4 d.p.)

- ☐ Equivalent to $X = 2$ or 3 with $X \sim \text{Bi}(5, 0.08704)$
- ☐ ≈ 0.0631

Answer / notes: $\hat{P}$ can only be 0.4 or 0.6 here, so sum $\Pr(X = 2) + \Pr(X = 3) \approx 0.0631$.

State average 1.0 / 2 (50%); 49% scored full marks. Mark distribution: 0: 45% · 1: 6% · 2: 49%

Your mark: _____ / 2

b.iv. Integer k making $\Pr(\text{late at least once in a week}) = 0.2$ (1 d.p.)

☐ Sets up $1 - \left(1 - \int_k^{59} f\right)^5 \approx 0.2$

☐ $k = 49$

Answer / notes: $k = 48$ gives 0.28, $k = 49$ gives 0.21 (rounds to 0.2), $k = 50$ gives 0.14. So $k = 49$.

State average $0.5 / 2$ (25%); 17% scored full marks. Mark distribution: 0: 67% · 1: 16% · 2: 17%

Your mark: _____ / 2

c.i. $\Pr(\text{wait} < 3.5)$ with $\mu = 2.5$, $\sigma = 0.6$ (2 d.p.)

☐ ≈ 0.95

Answer / notes: ≈ 0.95

State average $0.8 / 1$ (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: _____ / 1

c.ii. σ giving a 2% chance of waiting longer than 3.5 min (2 d.p.)

☐ $\sigma = \frac{1}{2.0537} \approx 0.49$

Answer / notes: $\frac{3.5-2.5}{\sigma} = 2.054 \Rightarrow \sigma \approx 0.49$

State average $0.5 / 1$ (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: _____ / 1

d. Probability distribution of Y , the number of red lights

☐ $\Pr(Y = 0) = 0.504$ and $\Pr(Y = 3) = 0.006$

☐ $\Pr(Y = 1) = 0.398$ and $\Pr(Y = 2) = 0.092$

Answer / notes: $0.8 \times 0.7 \times 0.9 = 0.504$. For $Y = 1$, add three cases ($0.126 + 0.216 + 0.056$). The rest follow.

State average $1.0 / 2$ (50%); 42% scored full marks. Mark distribution: 0: 41% · 1: 16% · 2: 42%

Your mark: _____ / 2

Your result

Total: _____ / 110 State average on these questions: **72 / 110 (65%)**

Topics to revise:

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